H524 Introduction to Biostatistics

AI-Enhanced Group Data Analysis Project

Fall 2025

Dr. John Molitor

Final Group Project Overview

Teams of 3-4 students will conduct a statistical analysis of an assigned medical dataset, with a **focus on comparing AI tool outputs**. This project replaces the traditional final exam and emphasizes critical evaluation of AI-assisted statistical analysis (20% of final grade).

1 Project Timeline

- **Week 3:** Team formation and dataset assignment
- **Week 4:** Check-in - Data familiarization meeting
- **Week 5-6:** AI tool comparison analysis phase
- **Week 7:** Draft analysis due for peer review (optional)
- **Week 8:** Final project submission
- **Week 9:** Group presentations (15 minutes + 5 minutes Q&A)

2 Learning Objectives

By completing this project, you will:

- Apply descriptive and inferential statistical methods to real health data
- Conduct appropriate hypothesis tests for your dataset
- **Compare outputs from three different AI tools** (Claude, ChatGPT, Microsoft Copilot)
- **Critically evaluate AI-generated statistical analyses**
- Document and verify AI-generated results
- Communicate statistical findings clearly and professionally

3 Dataset Assignments

Each team will be assigned ONE of the following datasets from the MedDataSets R package:

3.1 Dataset 1: Anorexia Treatment Study

- **File:** anorexia.csv
- **Sample Size:** 72 patients
- **Design:** Pre-post study with 3 treatment groups
- **Variables:** Treatment group (Control, CBT, Family Therapy), Pre-weight, Post-weight
- **Research Question:** Which treatment is most effective for weight gain in anorexia patients?
- **Methods:** ANOVA, paired t-tests, change score analysis
- **AI Divergence Expected:** High - Multiple valid analytical approaches

3.2 Dataset 2: Tooth Growth Study

- **File:** toothgrowth.csv
- **Sample Size:** 60 guinea pigs
- **Design:** 2×3 factorial (2 supplements $\times$ 3 doses)
- **Variables:** Tooth length, Supplement type (OJ vs VC), Dose (0.5, 1.0, 2.0 mg)
- **Research Question:** Does vitamin C promote tooth growth? Does delivery method matter?
- **Methods:** One-way or two-way ANOVA, pairwise comparisons
- **AI Divergence Expected:** High - Factorial design, interaction effects

3.3 Dataset 3: Low Birth Weight Study

- **File:** birthwt.csv
- **Sample Size:** 189 births
- **Design:** Observational case-control study
- **Variables:** Low birth weight (binary), mother's age, weight, race, smoking, hypertension
- **Research Question:** What factors are associated with low birth weight?
- **Methods:** Chi-square tests, t-tests, risk ratios
- **AI Divergence Expected:** High - Binary outcome, multiple predictors

3.4 Dataset 4: Pima Indian Diabetes Study

- **File:** pima_diabetes.csv
- **Sample Size:** 200 women
- **Design:** Observational cross-sectional study
- **Variables:** Diabetes status, pregnancies, glucose, BP, BMI, age
- **Research Question:** What factors predict diabetes in this population?
- **Methods:** T-tests, chi-square, comparison of means
- **AI Divergence Expected:** High - Data quality issues (missing data coded as 0!)
- **Special Note:** Tests AI data cleaning abilities

3.5 Dataset 5: Type 2 Diabetes Treatment Trial

- **File:** diabetes2.csv
- **Sample Size:** 699 adolescents
- **Design:** Randomized controlled trial
- **Variables:** Treatment (lifestyle, metformin, metformin+), Outcome (success/failure)
- **Research Question:** Which treatment is most effective for adolescent type 2 diabetes?
- **Methods:** Chi-square test, risk ratios, multiple comparisons
- **AI Divergence Expected:** Moderate - Cleaner dataset, but 3-group comparison

4 Required Analysis Components

4.1 Part 1: Descriptive Analysis (20 points)

- Data exploration and summary statistics
- Appropriate visualizations (3-4 high-quality plots)
- Check for missing data, outliers, and data quality issues
- Assessment of data distributions and assumptions

4.2 Part 2: Inferential Analysis (25 points)

- Formulate clear research question and hypotheses
- Select appropriate statistical test(s) based on data type and design
- Check and report assumption testing (normality, homogeneity of variance)
- Conduct hypothesis tests with proper interpretation
- Report effect sizes and confidence intervals
- Discuss practical vs. statistical significance

4.3 Part 3: AI Tool Comparison (30 points) — CRITICAL COMPONENT

This is the core learning objective of the project!

For your assigned dataset, you must:

4.3.1 Step 1: Consult Three AI Tools

Upload your dataset and ask for analytical guidance from:

1. **Claude** (claude.ai) - Upload CSV, ask for analysis
2. **ChatGPT** (chat.openai.com) - Upload CSV, ask for analysis
3. **Microsoft Copilot** (In RStudio) - Use for R code generation

Suggested prompt:

```
"I have a dataset on [brief description]. Please suggest
appropriate statistical methods to analyze [research question].
Check assumptions and provide R code for the analysis."
```

4.3.2 Step 2: Document AI Responses

Create a comparison table:

Aspect	**Claude**	**ChatGPT**	**Copilot**
Method Suggested			
Assumptions Checked			
Effect Sizes Reported			
Code Quality			
Interpretation Style			

4.3.3 Step 3: Critical Evaluation

Answer these questions in your report:

1. **Agreement:** Did all three AI tools suggest the same statistical method? If not, why did they differ?
2. **Rigor:** Which AI tool was most thorough in checking assumptions?
3. **Errors:** Did any AI tool make mistakes? What were they?
4. **Verification:** How did you verify the AI outputs were correct?
5. **Best Approach:** Which AI tool's approach did you ultimately follow? Why?
6. **Synthesis:** Did you create a "best of all three" hybrid approach?

4.3.4 Step 4: Verification Process

Document how you verified AI-generated results:

- Checked against course textbook and lecture notes
- Manually verified key calculations
- Ran AI-suggested code and examined output
- Consulted office hours or discussion board
- Cross-referenced with multiple AI tools

4.4 Part 4: Final Report and Presentation (25 points)

Written Report (5-6 pages):

- Introduction with research question (0.5 page)
- Methods: Data description and analytical approach (1 page)
- Results: Tables, figures, and statistical findings (2 pages)
- **AI Comparison Section** (1.5 pages) — Required!
- Discussion: Interpretation, limitations, conclusions (1 page)
- Appendix: AI usage log and R code

Group Presentation (15 minutes):

- Overview of dataset and research question (2 min)
- Key statistical findings (5 min)
- **AI tool comparison highlights** (5 min)
- Lessons learned about AI-assisted analysis (3 min)
- Q&A (5 min)

5 Deliverables

5.1 Check-In (Week 4) - See separate document

- Team meeting confirmation
- Dataset successfully loaded in R
- Preliminary descriptive statistics
- Initial data exploration

5.2 Final Submission (Week 8)

Submit via Canvas:

1. **Written Report (.pdf):** 5-6 pages following structure above
2. **R Script (.R):** Well-commented, reproducible code
3. **AI Comparison Document (.pdf):**
   - Screenshots of AI tool interactions
   - Comparison table
   - Critical evaluation
4. **Presentation Slides (.pdf):** 12-15 slides
5. **Dataset (.csv):** Your cleaned version (if modifications made)
6. **Team Contribution Statement:** Who did what

6 Grading Rubric

Component	Weight	Points
Descriptive Analysis	20%	20
Inferential Analysis	25%	25
AI Tool Comparison	30%	30
Report & Presentation	25%	25
Total	**100%**	**100**

6.1 Detailed Grading Criteria

6.1.1 Excellent (90-100%)

- Comprehensive and accurate statistical analysis
- **Thorough AI comparison with specific examples of differences**
- **Clear documentation of verification process**
- Professional presentation with compelling visualizations
- All team members contributed meaningfully

6.1.2 Good (80-89%)

- Mostly correct statistical analysis
- **Good AI comparison covering main differences**
- Clear presentation with good visualizations
- Evidence of verification, though not exhaustive

6.1.3 Satisfactory (70-79%)

- Basic statistical analysis with minor errors
- **AI comparison present but superficial**
- Adequate presentation meeting requirements
- Limited evidence of critical evaluation

6.1.4 Needs Improvement (¡70%)

- Major statistical errors or inappropriate methods
- **Minimal or no AI comparison**
- Poor presentation or missing components
- Little evidence of verification or critical thinking

7 Team Formation and Roles

Each team of 3-4 students should designate:

- **Data Manager:** Lead data cleaning and initial exploration
- **Statistical Lead:** Guide analytical approach and interpretation
- **AI Comparison Lead:** Coordinate AI tool consultations and comparison
- **Presentation Lead:** Coordinate slides and delivery

Note: All team members must contribute to all aspects. Roles are for coordination only.

8 Resources and Support

8.1 Technical Resources

- R and RStudio (required)
- AI Tools: Claude (claude.ai), ChatGPT (chat.openai.com), Microsoft Copilot
- MedDataSets R package (datasets also provided as CSV)
- Course textbook: Pagano & Gauvreau, *Principles of Biostatistics*

8.2 Getting Help

- Office Hours: Wednesday 10:00-11:30 AM
- Canvas Discussion Board for project questions
- Special project support sessions: Weeks 5, 6, 7
- Email for team meeting scheduling

8.3 Recommended R Packages

```
library(tidyverse)    # Data manipulation
library(ggplot2)      # Visualization
library(stats)        # Statistical tests
library(car)          # Assumption checks
library(effectsize)   # Effect size calculations
```

9 Academic Integrity

Important Reminders:

- AI-generated content must be verified and documented
- Each team member must contribute substantively
- Plagiarism will result in project failure
- Teams may discuss approaches, but analyses must be independent
- **You are responsible for correctness of all AI-generated content**

10 Tips for Success

1. **Start Early:** Don't wait until Week 7 to consult AI tools
2. **Meet Regularly:** Schedule weekly team meetings
3. **Document Everything:** Keep detailed notes of AI interactions
4. **Verify, Verify, Verify:** Never trust AI output without checking
5. **Ask Questions:** Use office hours if AI tools disagree
6. **Focus on Comparison:** The AI comparison is 30% of your grade!
7. **Practice Presentation:** Rehearse timing and transitions

11 What Makes This Project Different

Traditional Biostatistics Project:

- Analyze data
- Report results
- Done

This AI-Enhanced Project:

- Analyze data *using three different AI tools*
- Compare and contrast AI suggestions
- Critically evaluate which approach is best
- Verify and synthesize results
- **Learn to work WITH AI, not just use it**

This project prepares you for the reality of modern statistical practice, where AI tools are powerful assistants but require critical human oversight.